



SIMATS Online Programming Lab

C PROGRAMMING



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CO2 - AT2 - SCENARIO BASED QUESTIONS

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QUESTIONS

Q1

Scenario Based - Smart Waste Recycling Plant System
25 marks

**Q2**

Scenario Based - IT Company Project Performance Evaluation (2025–2026).
25 marks

**Q3**

Scenario Based - 3. Semiconductor Chip Manufacturing System
25 marks

**Q4**

Scenario Based - 4. Cloud Storage Subscription Management (2026)
25 marks



4/4 answered

Q1

Scenario Based - Smart Waste Recycling Plant System

25
marks

A recycling plant processes 5,000 kg of waste every day.

Waste Categories:

Plastic

Paper

Metal

Electronic Waste

Efficiency Levels:

≥90% Recycled → Excellent

75–89% → Good

50–74% → Average

Below 50% → Poor

(a) Develop a C program using arithmetic operators to calculate recycled waste percentage and remaining waste. Use decision-making statements to determine plant efficiency.

(b) Use looping constructs to process waste data for 30 days. Use continue to skip maintenance days and break if the processing machine fails.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     float plastic, paper, metal, ewaste;
4     float totalWaste, recycledWaste, recyc
```

```

5     int day;
6     printf("=== Smart Waste Recycling Plan\n");
7     printf("Enter recycled waste quantities\n");
8     printf("Plastic recycled (kg): ");
9     scanf("%f", &plastic);
10    printf("Paper recycled (kg): ");
11    scanf("%f", &paper);
12    printf("Metal recycled (kg): ");
13    scanf("%f", &metal);
14    printf("Electronic Waste recycled (kg): ");
15    scanf("%f", &ewaste);
16    totalWaste = 5000.0;
17    recycledWaste = plastic + paper + metal + ewaste;
18    recycledPercent = (recycledWaste / totalWaste) * 100;
19    remainingWaste = totalWaste - recycledWaste;
20    printf("\n--- Calculations (per day) ---\n");
21    printf("Total waste processed: %.2f kg\n", totalWaste);
22    printf("Total recycled waste: %.2f kg\n", recycledWaste);
23    printf("Recycled percentage: %.2f%%\n", recycledPercent);
24    printf("Remaining (not recycled): %.2f kg\n", remainingWaste);
25    if (recycledPercent >= 90.0) {
26        printf("Efficiency Level: Excellent\n");
27    } else if (recycledPercent >= 75.0 && recycledPercent < 90.0) {
28        printf("Efficiency Level: Good\n");
29    } else if (recycledPercent >= 50.0 && recycledPercent < 75.0) {
30        printf("Efficiency Level: Average\n");
31    } else {
32        printf("Efficiency Level: Poor\n");
33    }
34    printf("\n=== 30 Days Processing Loop\n");
35    for (day = 1; day <= 30; day++) {
36        if (day == 5 || day == 12 || day == 20 || day == 27) {
37            printf("Day %d: Maintenance day\n", day);
38            continue;
39        }
40        if (recycledWaste < 1000.0) {
41            printf("Day %d: Machine failed\n", day);
42            break;
43        }
44        printf("Day %d: Processing completed\n", day);
45    }
46    printf("\nProgram finished.\n");
47    return 0;
48 }

```



Input

stdin input



Output

Run

Save

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